

Solution Requirements

Date	
Team ID	
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	The system does not require user authentication. Users can directly access the application and perform HDI prediction without logging in, as the application is intended for public use and does not store personal user information	Low
2	Authorization levels	The system supports a single user role with equal access permissions. All users can select a country, enter HDI-related parameters, generate predictions, and view the prediction results. No rolebased access control is implemented.	Low

3	External interfaces	The system interacts with a preprocessed HDI dataset (CSV file), a trained Machine Learning model stored as a Pickle (.pkl) file, Flask web framework, HTML templates, and Pandas for data processing. No external APIs or third-party web services are required	Yes
		The system connects to a local CSV dataset, a trained Pickle (.pkl) Machine Learning model, the Flask backend, and HTML templates. No external APIs or cloud services are used.	
4	Transactions processing	The system accepts user inputs, validates the entered values, processes the feature data through the trained Linear Regression model, predicts the HDI score, classifies the predicted score into the appropriate HDI category, and displays the results to the user in real time.	Yes
5	Reporting	The system generates a prediction report displaying the selected country, input feature values, predicted HDI score, HDI category (Low, Medium, High, or Very High), and the trained model's performance (R^2 score) for user reference.	Yes

6	Business rules, etc.	The system validates all user inputs within predefined acceptable ranges. Missing or invalid values are replaced with dataset mean values where applicable. The predicted HDI score is restricted to the valid range (0-1) and classified according to predefined HDI threshold values	Yes
7	Compliance to laws or regulations	The system uses publicly available HDI datasets and does not collect or store personal user information. Since the application runs locally without external APIs or cloud services, no specific legal or regulatory compliance (such as GDPR or HIPAA) is required.	No
8	Other	The system supports automatic countrybased input pre-filling, robust input validation, local prediction using the trained model, error handling for invalid inputs, and a userfriendly web interface for displaying prediction results.	

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria

1	Performance & Speed	The application should process user input and generate HDI predictions quickly.	Prediction result displayed within 2 seconds .
2	Scalability	The application should support additional countries and larger datasets without affecting prediction accuracy.	Supports dataset expansion with 500+ countries/records
3	Security & Data Privacy	Supports dataset expansion with 500+ countries/records	Local processing with no personal data storage .
4	Reliability & Availability	Local processing with no personal data storage .	99% availability during local execution.
5	Usability & Accessibility	The system should provide a simple and intuitive web interface with country selection and manual input options.	Easy-to-use interface with minimal user training required.
6	<i>Other</i>	The application should be easy to maintain, update, and deploy on different operating systems supporting Python and Flask.	Compatible with Windows, Linux, and macOS using Python 3.x and Flask.